RPL

IPv6 Routing Protocol for LLNs

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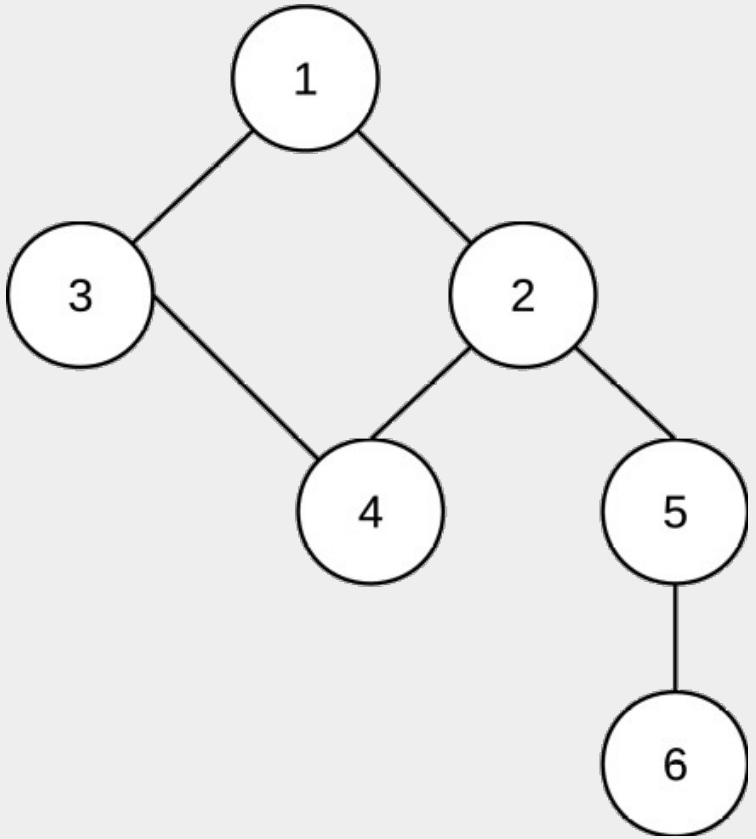
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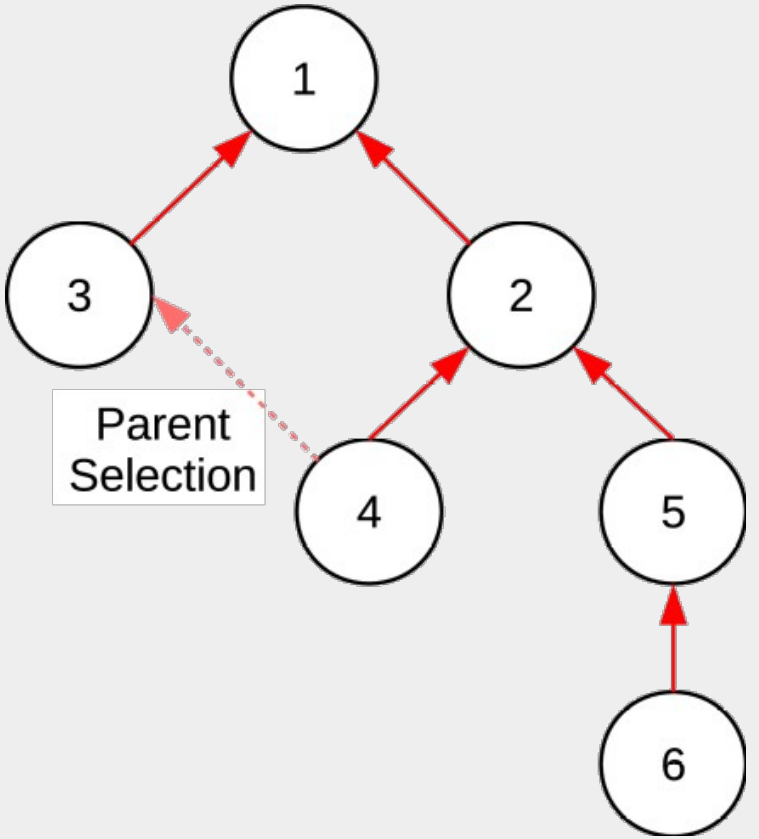
What's RPL?

- IPv6 Routing Protocol for Low-Power and Lossy Networks
- IETF Routing Protocol – ROLL WG
- Simplification of the routing protocol
  - Tree like topology in a mesh
    - Root node
    - Nodes have one parent
    - Nodes can have n-childs
  - Out of scope
    - Self-healing
    - Floating root nodes
    - Security, a lot of more...

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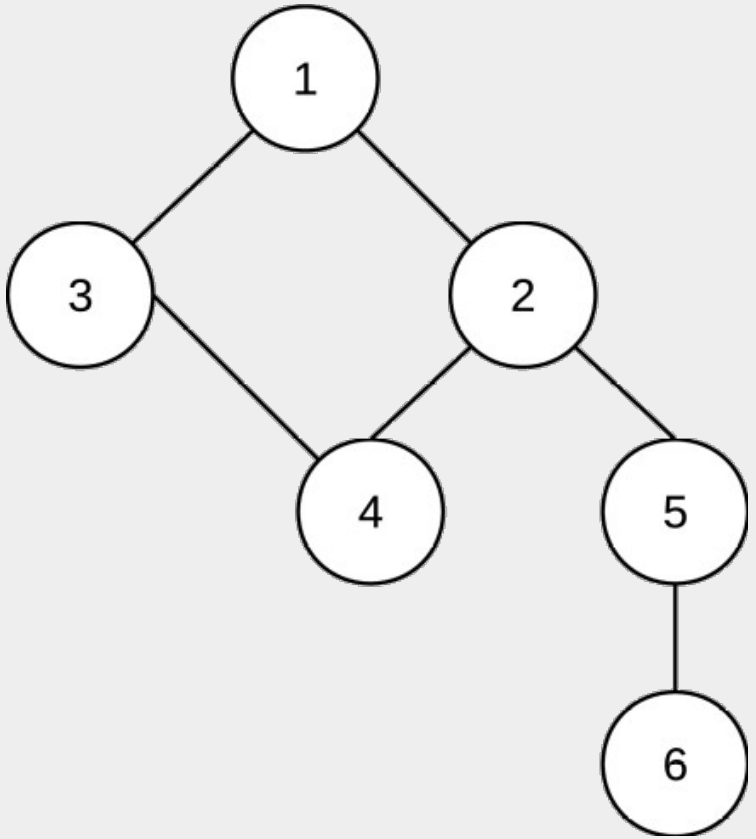
Upward Routes
storing and non-storing

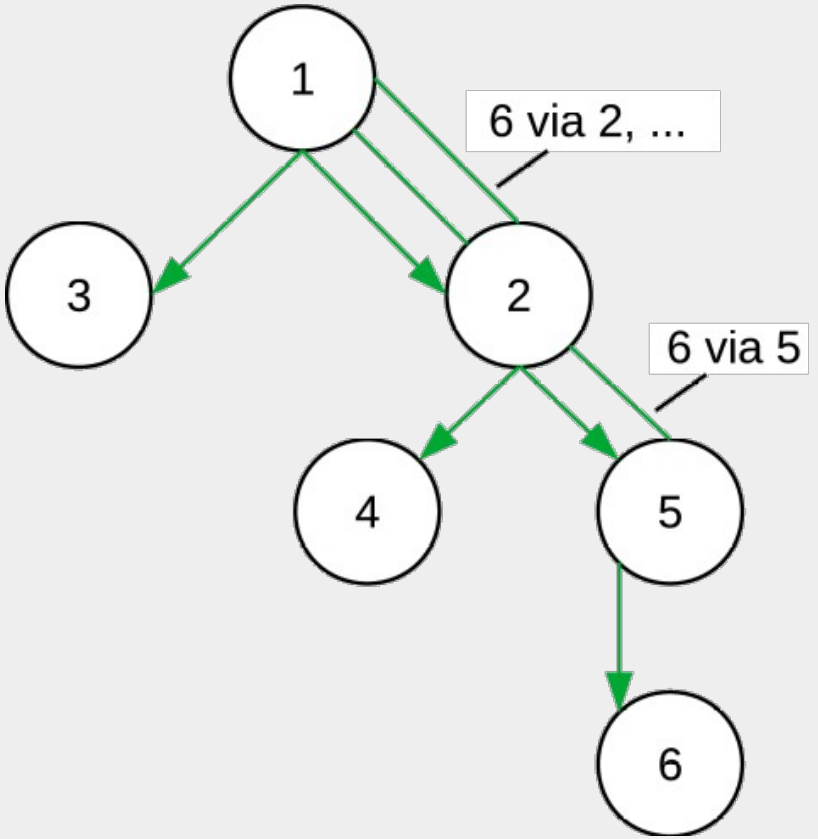




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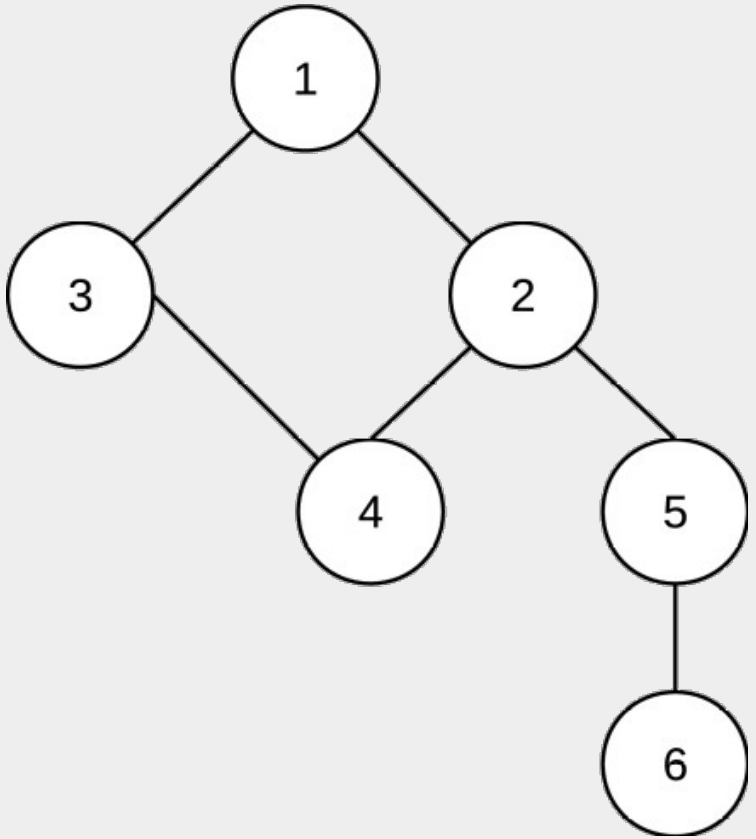
Downward Routes storing

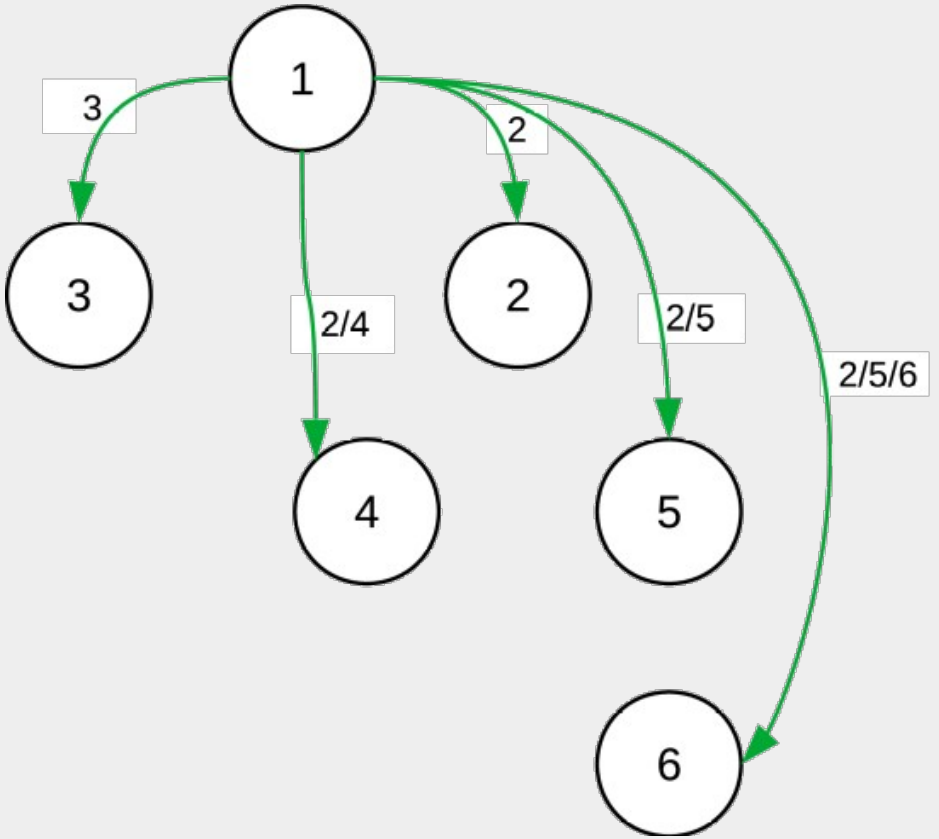




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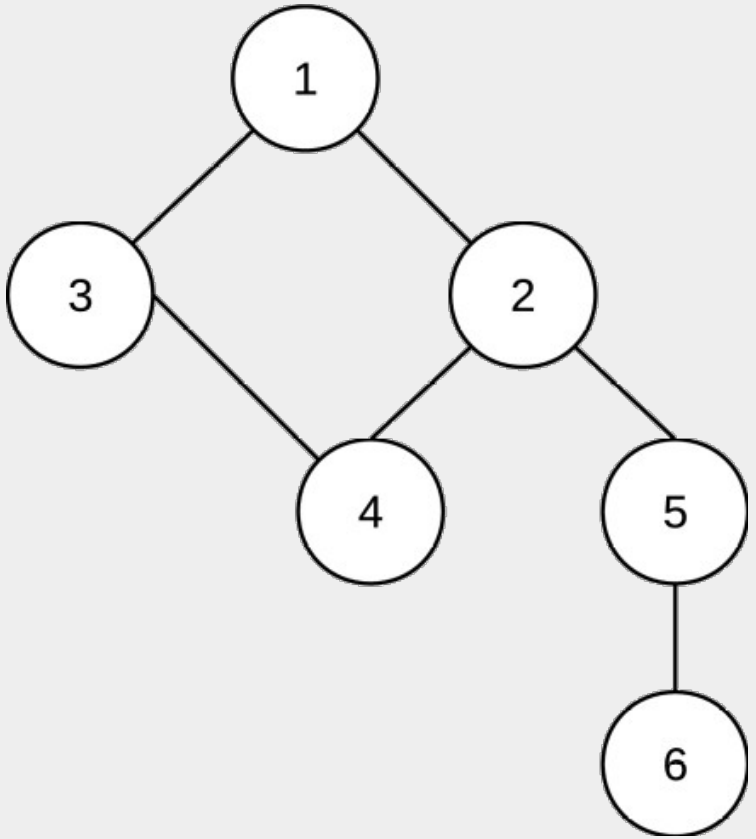
Downward Routes non-storing

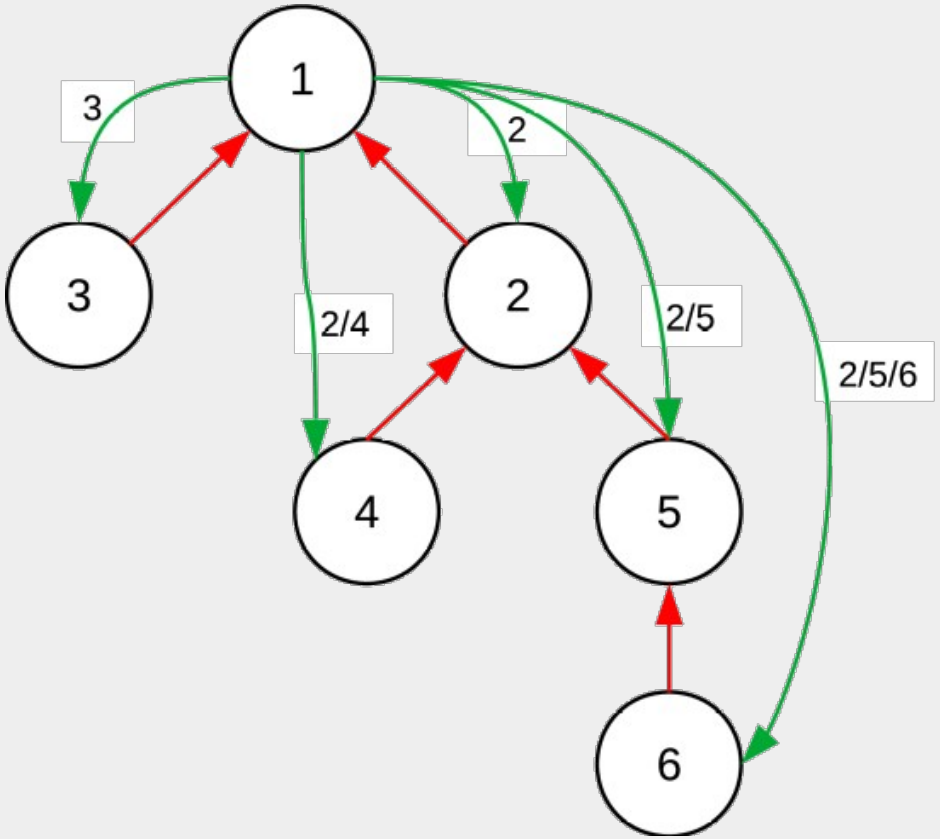




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Downward and Upward Routes non-storing





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RPL – Source Routing Header

- Only Root Node inserts source routing extension header
- IPv6 only
- Forwarding
  - Address swapping
  - Loop detection
- Compression
  - Addresses inside the segment array
  - According to destination address

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Implementation
Kernel

- Leightweight Tunnel (net/ipv6/rpl_iptunnel.c)
  - Like IPv6 Segment Routing (the other source routing)
    - Per route prefix and adding source routing header
    - Cannot setup IP encapsulation yet
  - Config: IPV6_RPL_LWTUNNEL
- Forwarding (net/ipv6/exthdrs.c)
  - Always build into IPv6
  - Disabled by default (can be enabled via sysfs)
- Compression Helpers (net/ipv6/rpld.c)

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Implementation
rpld

- Experimental user space daemon
- Exchange ICMPv6 messages
- Root node stores tree data structure to represent topology
- Non-storing mode operation

  1) Creating upward routes
  2) Nodes report parent and own address to root node
  3) Root node setup RPL source routing entries

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DEMO

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Future Work

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